Homework 5

Due: Friday, October 9 at 11:59 p.m. PT

MATH 112, Fall 2026

Instructions

- Write your full name, "Homework 5," and the date at the top of the first page.
- Show all work and explain your reasoning. Write in complete sentences.
- Typeset your solutions in LaTeX if you can.
- Submit a single PDF file to Gradescope under the assignment "Homework 5."
- Use Gradescope to electronically match problems to pages in your PDF.
- Questions? Email the instructor or come to office hours; see Moodle.
- You are strongly encouraged to work together. Write up your own solutions.

Be sure to show your work. Organize it so that another MATH 112 student who has not seen the problems can follow your reasoning from line to line. Solutions that contain only an answer, without justification or supporting work, will not receive full credit.

Problem 1 (12 points)

Let A, B be two nonempty bounded subsets of $\mathbb{R}$.

1. Prove that $A \cup B$ is bounded and $\sup(A \cup B) = \max\{\sup A, \sup B\}$, $\inf(A \cup B) = \min\{\inf A, \inf B\}$. Can we say something analogous about $\sup(A \cap B)$ or $\inf(A \cap B)$?
2. If $\sup A = \inf B$ prove that for every $\varepsilon > 0$ there exist $a \in A$ and $b \in B$ such that
$$b - a < \varepsilon.$$
3. If $A \subseteq B$, prove that
$$\inf B \leq \inf A \leq \sup A \leq \sup B.$$

Problem 2 (10 points)

1. Let A, B be nonempty subsets of $\mathbb{R}$ with A upper bounded and B lower bounded. Prove that $\sup A \leq \inf B$ if and only if, for every $a \in A$ and every $b \in B$, $a \leq b$.
2. Let A, B be nonempty subsets of $\mathbb{R}$ that are upper bounded and have the following property: for every $a \in A$, there exists $b \in B$ such that
$$a \leq b.$$

Prove that
$$\sup A \leq \sup B.$$

Problem 3 (16 points)

Find, if they exist, the maximum, minimum, supremum, and infimum of each of the following sets in $\mathbb{R}$.

(a) $A = \{x \in \mathbb{R} : x > 0, 0 < x^2 - 1 \leq 2\}$,
(b) $B = \{x \in \mathbb{Q} : x \geq 0,\ 0 < x^2 - 1 \leq 2\}$
(c) $C = \left\{0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$.
(d) $E = \left\{\frac{1}{n} + (-1)^n : n \in \mathbb{N}\right\}$
(e) $G = \left\{5 + \frac{6}{n} : n \in \mathbb{N}\right\} \cup \{7 - 8n : n \in \mathbb{N}\}$.
(f) $H = \left\{1 + (-1)^n + \frac{(-1)^{n+1}}{n} : n \in \mathbb{N}\right\}$,
(g) $I = \left\{\frac{1}{2^n} + \frac{1}{3^m} : n, m \in \mathbb{N}\right\}$.
(h) $J = \left\{\frac{(-1)^n m}{n+m} : n, m \in \mathbb{N}\right\}$